

Mirroring in Microsoft Fabric

SQL Server Edition

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Sponsors & Organizers



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About me

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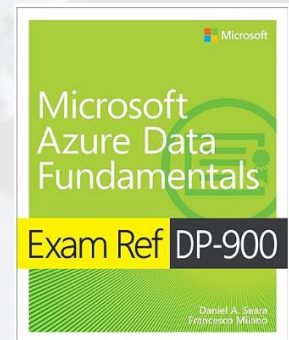
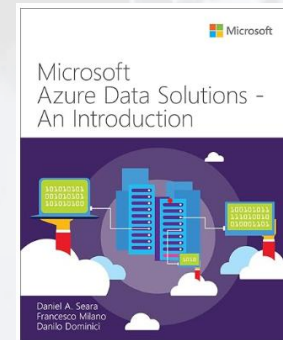
Working with SQL Server since 2005 edition

Working with Azure Data Platform since 2012

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AGENDA

Agenda

1

Why Mirroring

The gap between
operational data
and analytics.

2

Architecture

What Fabric
creates and how
data reaches
OneLake.

3

SQL Server paths

2016–2022 versus
2025, and how to
choose.

4

Demo

Setup, change,
observe, query.

5

Operations & limits

Monitoring,
schema changes,
security,
boundaries.

1. Why Mirroring

2. Architecture

3. SQL Server paths

4. Demo

5. Operations & close

The operational analytics gap

01

Yesterday's extract

Reporting still refreshes nightly or hourly, so decisions run on data that is already old.

02

Today's decisions

The business asks for operational answers now, not after the next batch window.

03

OLTP must stay healthy

Analytical queries cannot be allowed to compete with the production transactional workload.

Fabric in 60 seconds

01

OneLake: one logical data lake

A single storage foundation for the whole tenant, in open Delta format.

02

Multiple engines, shared data

SQL, Spark and Power BI read the same copy instead of duplicating it.

03

Power BI built in

Reporting is part of the platform, not a separate downstream stack.

Mirroring: the high-level flow

1

Initial snapshot

Eligible source tables are materialized once as a starting baseline.

2

Continuous changes

Inserts, updates and deletes are applied continuously after the seed.

3

Analytics-ready copy

Fabric keeps Delta tables in OneLake; the source stays the system of record.

Where Mirroring exists in Fabric

Azure SQL Database

Fully managed PaaS source, mirrored directly from the service.

Azure SQL Managed Instance

Same pattern for instance-scoped workloads in Azure.

SQL Server — our focus today

On-premises and hybrid servers, including Azure Arc-enabled machines.

Cosmos DB · Snowflake · Oracle · others

Context only: the supported list evolves, so check Microsoft Learn.

End-to-end architecture

1

SQL Server

Transactions commit on the operational database as usual.

2

Capture + secure connectivity

Changes are exposed and transported to the Fabric mirroring service.

3

OneLake Delta tables

Fabric lands and maintains the analytical representation.

4

SQL · Spark · Direct Lake

Consumers query Fabric, never the production database.

What Fabric creates

Mirrored database item

Owens replication state and table selection.

Replicated tables in OneLake

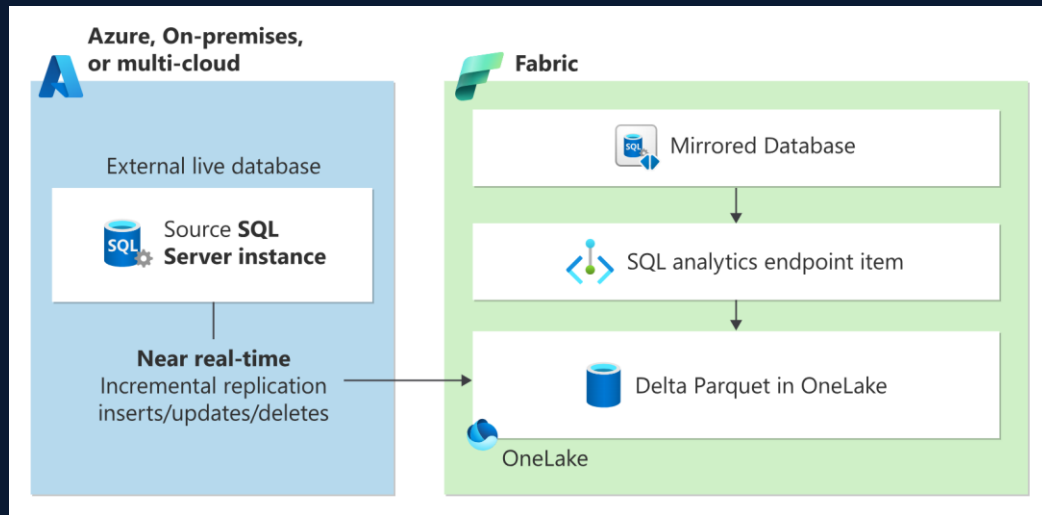
Delta tables kept current by the service.

Read-only SQL analytics endpoint

A T-SQL surface for analytical queries.

Semantic model with Direct Lake

Modeling remain your responsibility.



Near real time, not instantaneous

1

Commit

Transaction shape and duration start the clock.

2

Capture

Change volume and capture health drive the first delay.

3

Transfer

Gateway or Arc connectivity moves the changes.

4

Apply

Fabric writes into the Delta tables.

5

Query

Endpoint and semantic model add the final hop.

The version decision

Which SQL Server version is the source?

PATH A

SQL Server 2016–2022 → CDC-based path

Change data capture is the prerequisite, and the on-premises data gateway carries the changes.

PATH B

SQL Server 2025 → Fabric mirroring + Azure Arc

A purpose-built mirroring capability integrated with Azure Arc-enabled SQL Server.

Path A — SQL Server 2016–2022

CDC captures row changes

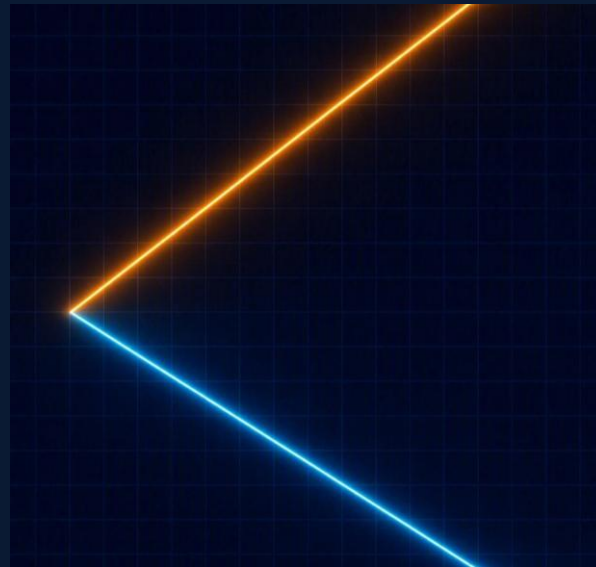
Inserts, updates and deletes are exposed by the capture process.

Fabric reads through the gateway

The on-premises data gateway bridges the private network.

DBA owns CDC and log health

Capture latency, cleanup and log reuse become day-two work.



Path B — SQL Server 2025

Native Fabric mirroring capability

Change feed built into the engine, no CDC prerequisite to maintain.

Azure Arc-enabled SQL Server

The server is projected into Azure and the extension is configured there.

Managed identity-based integration

Identity, outbound connectivity and supported builds still need planning.

Same destination, different source mechanics

2016-2022

CDC + gateway

Enable CDC, run a gateway, and own capture and log health.

2025

Native feed + Azure Arc + gateway

Onboard to Arc, enable the capability, and grant the identity. Data Gateway still needed.

Both land the same OneLake analytical copy

The decision inputs are your installed version, CDC and gateway operations, Arc adoption and schema-change process.

Prerequisites — Fabric side

01

Fabric capacity and workspace

The workspace must sit on an eligible capacity.

02

Tenant settings enabled

Admins must allow Fabric items and the mirroring capability.

03

Member/Admin to create the item

Creators need sufficient workspace rights.

04

Supported region and source

Check region availability on Learn before you plan.

Prerequisites — SQL Server side

01

Supported version, edition, build

Verify the exact patch level, not just the major release.

02

Eligible user tables

Inventory data types, special features and existing CDC.

03

Stable primary keys where required

Key design decides whether a table can be mirrored.

04

Least-privilege login / identity

Use a dedicated principal when possible

Connectivity in one slide

Gateway for the CDC path

It bridges Fabric and a private SQL Server.

Outbound secure connections

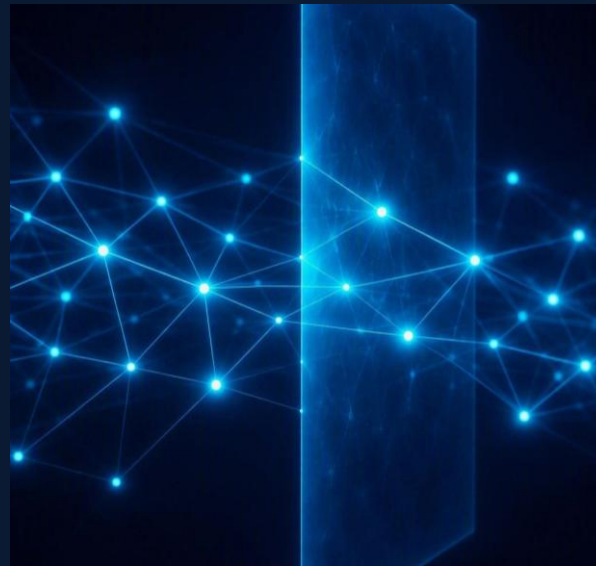
Keep inbound exposure minimized.

Place the gateway near the source

Network distance shows up as latency.

Design gateway resiliency

Treat it as production infrastructure.



Identity and permissions

01

Source principal

Grant only the permissions Microsoft lists for reading changes.

02

Fabric workspace

Least privilege for who administers the mirrored item.

03

Consumer access

Endpoint and semantic-model security is a separate design.

Configure Path A — 2016–2022

1

Validate & enable CDC

Confirm the capture process runs as documented.

2

Configure login and gateway

Test that the credential reaches the database.

3

Create mirrored database

Choose SQL Server and the documented connection.

4

Select tables and start

Begin with a coherent subject area.

CDC health

01

CDC enabled?

Database and table level, verified — not assumed.

02

Capture job healthy?

Watch capture latency and cleanup behaviour.

03

Log reuse healthy?

Blocked truncation turns into storage pressure.

04

Long transactions?

They delay capture and amplify log growth.

Configure Path B — SQL Server 2025

1

Arc-enable SQL Server

Project the server into Azure as an Arc resource. Data Gateway still needed.

2

Enable Fabric mirroring

Configure the capability per Microsoft guidance.

3

Grant identity access

Wire up the managed identity and permissions.

4

Create mirror in Fabric

Then select tables and start replication.

Table selection: start intentionally

Mirror only operationally valuable tables

Start with one coherent subject area: shorter seeding, smaller blast radius.

Check keys, types and special features

Temporal, encrypted, memory-optimized and LOB patterns need verification.

Baseline row counts and change rate

Record sizes, daily change volume and release cadence before you start.

DEMO

SQL Server 2019

Setup → Change → Observe → Query

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Monitor at two levels

Database state

Running, seeding, warning, stopped or failed.

Per-table state

A green item does not prove every table is current.

Replication progress / latency

Track freshness against your business target.

Source health

Capture, log reuse, CPU, I/O and connectivity.



Schema changes are an operational event

Test DDL behaviour

Supported behaviour differs by path and by DDL operation.

Know when a table reseeds

Some changes require intervention or a full reinitialization.

Coordinate application releases

Add a mirroring compatibility check to the release procedure.

Stop, restart, and reseed

Stopping is not just pausing a dashboard

Lifecycle actions can affect retained replication state.

Restart can require reinitialization

A new initial load may be triggered.

Plan time and source impact

Estimate reseed duration and network cost, and keep a runbook.

Security does not mirror itself

RLS, masking and permissions need design

Source policies are not automatically enforced for Fabric consumers.

Sensitive columns still arrive as data

Classify before you select tables.

Secure endpoint and semantic model

Rebuild and test the access model with real personas.



Current boundaries to validate

01

Table and object eligibility

Not every object can be mirrored.

02

Data types and large values

LOB and special types need checking.

03

Table-count and scale limits

Verify current numbers on Learn.

04

DDL and reseed behaviour

Some changes force reinitialization.

05

HA topology support

Test listener and failover behaviour.

06

No cross-tenant assumption

Plan tenancy and region explicitly.

Troubleshooting: isolate the layer

1

Source capture

Confirm new changes are capturable at the source.

2

Identity & connectivity

Validate the gateway or Arc connection and credential.

3

Fabric replication

Inspect item and table errors.

4

Consumer surface

Test endpoint and semantic model separately.

What Mirroring costs in Fabric

01

Mirroring compute is free

Replication into OneLake consumes no Fabric capacity.

02

Mirrored storage is free

Free OneLake storage up to 1 TB per capacity unit.

03

You pay only when you query

SQL endpoint and Power BI use normal capacity.

04

The real cost sits on the source

CDC overhead, network and SQL Server licensing.

THANK YOU

Questions?

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Useful resources

Mirroring in Microsoft Fabric — overview

<https://learn.microsoft.com/en-us/fabric/database/mirrored-database/overview>

Mirroring for SQL Server in Fabric

<https://learn.microsoft.com/en-us/fabric/mirroring/sql-server>

Configure SQL Server mirroring – 2016-2022

<https://learn.microsoft.com/en-us/fabric/mirroring/sql-server-tutorial?tabs=sql201622>

Configure SQL Server mirroring – 2025

<https://learn.microsoft.com/en-us/fabric/mirroring/sql-server-tutorial?tabs=sql2025>

Monitor a Fabric mirrored database

<https://learn.microsoft.com/en-us/fabric/database/mirrored-database/monitor>

SQL Server mirroring limitations

<https://learn.microsoft.com/en-us/fabric/mirroring/sql-server-limitations>